THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The number of electrons transferred in the formation of Al_2O_3 in total is: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 6

2. 4.8 g of Mg burns in oxygen to form 8 g of MgO. The mass of oxygen used is: 1

(a) 12.8 g **(b)** 3.2 g **(c)** 4.8 g **(d)** 8 g

3. The formula unit mass of K_2SO_4 is (K = 39, S = 32, O = 16): 1

(a) 174 u **(b)** 136 u **(c)** 126 u **(d)** 110 u

4. An ionic compound dissolves in water and the solution conducts electricity because: 1

(a) atoms are free **(b)** molecules are charged

(c) ions are free to move **(d)** electrons are free

5. Percentage of hydrogen in water is: 1

(a) 11.1 per cent **(b)** 5.5 per cent **(c)** 88.9 per cent **(d)** 33.3 per cent

6. The formula unit mass of Al_2O_3 is (Al = 27, O = 16): 1

(a) 75 u **(b)** 86 u **(c)** 102 u **(d)** 43 u

7. Which of these has the lowest melting point? 1

(a) MgO **(b)** CaF_2 **(c)** CH_4 **(d)** $NaCl$

8. The unified atomic mass unit (u) is defined as: 1

(a) 1/16 of the mass of an oxygen molecule **(b)** 1/12 of the mass of one carbon-12 atom

(c) the mass of 12 atoms of carbon **(d)** the mass of one hydrogen atom exactly

9. The molecular mass of glucose $C_6H_{12}O_6$ is (C = 12, H = 1, O = 16): 1

(a) 180 u **(b)** 360 u **(c)** 90 u **(d)** 162 u

10. Brittleness of ionic crystals is due to: 1

(a) weak bonds

(b) repulsion between like ions when layers shift

(c) free electrons

(d) molecules sliding

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State two differences between a molecule and a formula unit. 2

12. Write the formula for
(a) potassium nitrate
(b) zinc sulphate. 2

13. Why does the mass of an open container appear to decrease when a reaction produces a gas? 2

14. Name the ions in
(a) $CaCl_2$
(b) Na_2SO_4. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Calculate the molecular mass of C_2H_6, C_2H_4 and C_2H_2 (C = 12, H = 1). 3

16. Explain why 2 g of hydrogen and 32 g of oxygen produce 36 g of water by the law of constant proportions with leftover. 3

17. Distinguish between molecular mass and formula unit mass with examples. 3

OR

The percentage composition of a compound is 40 per cent carbon, 6.7 per cent hydrogen and 53.3 per cent oxygen. Show that the ratio is consistent with a molecular mass of 180 u. 3

18. Explain Dalton's postulate that atoms combine in simple whole number ratios with examples of CO and CO2. 3

OR

Explain why the law of conservation of mass is verified in a closed container for the reaction of vinegar and baking soda. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Calculate the mass percentage of each element in $NaNO_3$ and $(NH_4)_2SO_4$ (Na 23, N 14, O 16, H 1, S 32). 5

OR

Explain the formation of the covalent compounds HCl, H_2O and NH_3 by sharing of electrons, giving the number of shared pairs and the number of lone pairs on the central atom. 5

20. Explain why different samples of water from different places have the same composition but sea water and river water differ. Use the terms compound and mixture. 5

OR

Describe in detail the experiment that verifies the law of conservation of mass and the sources of errors that may arise. 5

END OF QUESTION PAPER — Total: 40 Marks